When crop rankings do not determine land allocations: set-valued reversal under operational and downside-risk constraints

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Supervisor-review draft

Abstract

Crop rankings compress heterogeneous agronomic and economic information into an ordering, but land allocation is a cardinal, constrained portfolio decision. We formalize the gap between these objects in a multi-crop stochastic programme with operational constraints and loss-based conditional value at risk. The framework treats the optimum as a set and distinguishes possible, universal and selected ranking reversal. It establishes that an ordinal ranking alone cannot identify acreage, provides sufficient conditions for feasibility-forced reversal, and replaces a presumed unique dependence threshold with potentially disconnected crossing sets. We connect these results to an auditable analysis of corn, soybeans and winter wheat in 26 US states over 2022–2024. Operating-margin and acreage top ranks differ in 63 of 77 state-years (81.8%), although rates vary materially across four ranking definitions and national ranks agree in all 9 crop-year comparisons. A preregistered simulation maps potential mechanisms but fails its precision gate and is therefore not headline evidence. Together, the results show that ranking-to-acreage disagreement is empirically visible while its optimizing mechanism remains unidentified. Rankings should consequently be evaluated as inputs to an explicit feasible decision problem, not interpreted as allocations by themselves.

1 Introduction

Crop prediction systems increasingly turn environmental and management data into yield forecasts, suitability scores and crop rankings [Liakos et al., 2018, van Klompenburg et al., 2020]. These outputs are useful summaries, but they do not answer the operational question of how much land to assign to each crop. Acreage decisions must trade expected returns against land availability, budgets, rotations, contracts and joint downside exposure. Treating the highest-ranked crop as the crop that should receive the most land therefore crosses an unexamined boundary from ordinal prediction to cardinal prescription.

Crop planning research has long represented land use as a constrained portfolio problem. Models incorporate resource constraints, stochastic prices and yields, rotations, robustness and downside risk [Filippi et al., 2017, Boyabatlı et al., 2019, Benini et al., 2023, Randall et al., 2022, 2024]. In parallel, prescriptive analytics and decision-focused learning show that predictive accuracy and downstream decision quality are distinct evaluation targets [Bertsimas and Kallus, 2020, Elmachtoub and Grigas, 2022, Wilder et al., 2019, Mandi et al., 2023]. Yet a basic interpretive question remains: what, exactly, can be inferred when a recommendation ranking and an observed or optimized allocation disagree?

This question is difficult for three reasons. First, a ranking is unchanged by every strictly increasing transformation of its scores, whereas proportional allocations are not. Second, constrained optimization may have multiple optimal solutions, so a reversal can hold for one selected optimizer, at least one optimizer or every optimizer. Third, dependence affects portfolio losses through the complete joint distribution; a scalar tail-dependence coefficient cannot generally order conditional losses or acreage

[Demarta and McNeil, 2005, Ansari and Rockel, 2024]. These distinctions invalidate simple equivalences between ranks, marginal optimality conditions and acreage levels.

Here we retain the teacher Draft's integrated multi-crop architecture but repair its claims around a set-valued identification framework (Fig. 1). We formulate expected-profit maximization over a polyhedral operational set under loss-CVaR, using the atom-safe representation of Rockafellar and Uryasev [2000, 2002]. We define possible, universal, selected, strong and top-rank reversal; derive a complete subgradient Karush–Kuhn–Tucker characterization; and describe dependence effects through restricted distributional ordering and crossing sets rather than a universal unique threshold. Information and flexibility are retained through actionability and weak feasible-set monotonicity, not asserted to be strictly complementary in general.

The empirical analysis then asks a deliberately narrower question: how often do transparent rankings and observed planted-acreage ranks disagree in an official complete-case state sample? Across 77 three-crop state-years in 26 states, operating-margin top ranks disagree with acreage ranks in 63 cases. This pattern is definition-sensitive, with top-rank discordance ranging from 62.3% to 87.0%, and it disappears in the retained national comparison. These are descriptive facts, not evidence that observed acreage solves our stochastic programme or that CVaR, copulas or any single constraint caused the disagreement.

The paper contributes a disciplined bridge between rankings and allocations. It establishes what ordinal information cannot identify, supplies solution-set-aware reversal definitions, integrates operational and downside-risk constraints in a reproducible model, and demonstrates how empirical discordance can be reported without assigning an unobserved mechanism. The simulation component is also informative by failure: although all primary solves and numerical replays pass, none of the predeclared convergence rows meets the joint precision rule. We therefore quarantine its substantive patterns to the Supplementary Information. This separation of theorem, simulation, empirical and boundary evidence is central to the study.

2 Related literature

Prediction and prescription in agriculture. Agricultural machine learning commonly predicts yields or classifies agronomic outcomes from weather, soil, sensor and remote-sensing inputs [Liakos et al., 2018, van Klompenburg et al., 2020]. Such models can support a ranking, but they do not encode the feasible set or objective required for acreage prescription. The broader prescriptive-analytics literature evaluates predictions through downstream decisions and regret [Bertsimas and Kallus, 2020, Elmachtoub and Grigas, 2022], while decision-focused learning differentiates or searches through optimization structure [Wilder et al., 2019, Mandi et al., 2023]. Our analysis does not propose a new learning algorithm. It instead isolates the identification gap that must be resolved before a prediction can be interpreted as a land allocation.

Constrained crop planning. The closest included crop-planning studies optimize crop mix under stochastic returns, rotations or climate uncertainty. Filippi et al. [2017] combine realistic crop-selection constraints with a CVaR criterion; Boyabatlı et al. [2019] link allocation and rotation benefits dynamically; Benini et al. [2023] formulate multi-period rotation and diversification constraints; and Randall et al. [2022] study robustness across climate projections. The systematic review by Randall et al. [2024] documents the diversity of optimization approaches in the field. These studies motivate a multi-crop operational set, but they do not equate ordinal recommendations with the entire optimal solution set.

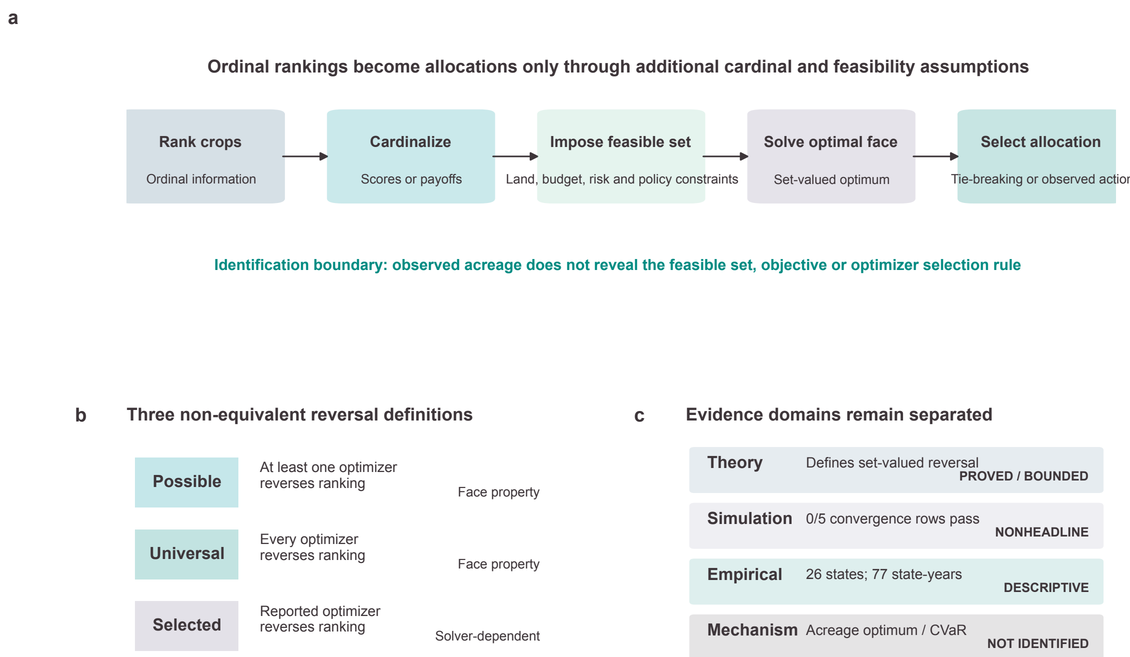

Figure 1: From ordinal crop rankings to set-valued allocation claims. The flow distinguishes additional cardinal and feasibility assumptions, three non-equivalent reversal definitions, and the boundaries between theorem, simulation and empirical evidence.

Downside risk and dependence. Loss-CVaR is convex and admits a finite-scenario auxiliary representation that remains valid with atoms [Rockafellar and Uryasev, 2000, 2002]. Copula models can separate marginal distributions from dependence, but tail properties vary by family [Demarta and McNeil, 2005, Ansari and Rockel, 2024]. We consequently condition any dependence comparison on a named family and a verified ordering of portfolio-loss distributions. This is stricter than treating a lower-tail coefficient as a universal mechanism.

Information and operational flexibility. Stochastic programming separates here-and-now choices from actions taken after information arrives [Dantzig, 1955]. Information has operational value only when it changes feasible decisions [Keisler, 2004, Merkhofer, 1975]; the value of flexible resources likewise depends on margins, costs and the complete uncertainty environment [Van Mieghem, 1998]. These results support non-negativity and weak action-set monotonicity, but not general strict complementarity between signal precision and flexibility.

Relative to this verified set, the present study differs by jointly comparing an ordinal crop ranking with an entire constrained loss-CVaR optimal set, separating possible, universal and selected reversals, and pairing that theory with a claim-bounded descriptive empirical audit. This is a bounded distinction, not a claim to be the first study of crop allocation, CVaR or prediction-informed decisions.

3 Integrated multi-crop stochastic model

Let crops be indexed by $i = 1, \dots, n$ and let x_i denote acreage. Random per-acre profit is $\widetilde{\pi}_i = \widetilde{p}_i \widetilde{y}_i - c_i$, where price and yield may be dependent within and across crops and baseline cost c_i is deterministic. Portfolio profit and loss are

$$\Pi(x, \omega) = \sum_i x_i \widetilde{\pi}_i(\omega), \qquad L(x, \omega) = -\Pi(x, \omega). \tag{1}$$

Returns are linear in acreage; fixed setup costs, crop interactions and endogenous price feedback are outside the baseline model.

Operational feasibility is represented by the non-empty polytope

$$X = \{x \in \mathbb{R}^n : \ell \le x \le u,\ \mathbf{1}^\top x \le A,\ c^\top x \le B,\ Gx \le g\}. \tag{2}$$

The rows of $Gx \le g$ accommodate rotation, contract, agronomic and other shared restrictions. Land is an inequality because idling may be optimal or forced; full use requires an admissible profitable expansion. Bounds, budgets and shared restrictions in the simulation are illustrative stresses, not estimated private farm constraints.

For confidence level $\alpha \in (0, 1)$, loss-CVaR is

$$r_\alpha(x) = \mathrm{CVaR}_\alpha[L(x)] = \min_{v \in \mathbb{R}} \left\{ v + \frac{1}{1-\alpha} \mathbb{E}[(L(x) - v)_+] \right\}. \tag{3}$$

The canonical problem maximizes expected profit $\mu^\top x$, where $\mu_i = \mathbb{E}[\widetilde{\pi}_i]$, subject to $x \in X$ and $r_\alpha(x) \le \kappa$. The sign convention protects against large losses. For finite scenarios s with positive weights w_s, the problem is a linear programme after introducing a free v and non-negative excess-loss variables q_s.

Marginal profit distributions F_i and a valid n-copula C_θ define the joint law. Pairwise lower-tail dependence $\lambda_{L,ij}$ is a diagnostic when its limit exists, but it does not identify a copula, a conditional tail mean or an allocation. We therefore vary named Gaussian, Student-t and Clayton families in simulation and make no cross-family scalar ordering claim.

A suitability vector s induces a weak order; ties form equivalence classes. Proportional and winner-take-all rules are benchmark mappings and are not feasible optima by definition. Let

$$S_\kappa = \arg\max\{\mu^\top x : x \in X,\ r_\alpha(x) \le \kappa\}. \tag{4}$$

For a strict pair $s_i > s_j$ and tolerance τ_x, reversal at x means $x_j - x_i > \tau_x$. Reversal is *possible* if it holds at some $x \in S_\kappa$, *universal* if it holds at every $x \in S_\kappa$, and *selected* if it holds for a declared deterministic selection. Strong and top-rank reversal add zero-allocation and highest-tie-class conditions, respectively.

4 Repaired structural results

The first result is an identification statement. A weak or strict suitability ordering alone does not determine a cardinal acreage vector. Any monotone transformation preserves the ordering but may alter a proportional rule, and neither the transformation nor the ordering supplies an objective, constraints or optimizer selection. This insufficiency is not a numerical failure; it follows from the mismatch between ordinal and cardinal objects.

Existence and tractability follow under conventional conditions. If X is non-empty and compact, per-acre profits are integrable and the risk-feasible set is non-empty, an optimizer exists. The risk-feasible set is convex, and the finite-scenario problem is linear. These conditions preserve the multi-crop stochastic architecture while making its domain explicit.

Operational constraints can force a universal reversal without a risk mechanism. If $s_i > s_j$ and

$$\sup_{x \in X} x_i + \tau_x < \inf_{x \in X} x_j, \tag{5}$$

then every feasible allocation reverses the pair. The bound condition $u_i + \tau_x < \ell_j$ is sufficient. Conversely, if an unconstrained expected-profit optimum is risk-feasible, the risk-constrained optimal value is unchanged and the constrained solution set is the intersection of the unconstrained optimal set with the risk-feasible set. One observed slack optimum does not establish equality of solution sets when optima are multiple.

The complete first-order characterization also blocks an acreage-order shortcut. At an optimum x^*, there are non-negative multipliers and a subgradient $d \in \partial r_\alpha(x^*)$ such that

$$0 = -\mu + \lambda d + \gamma \mathbf{1} + \beta c + G^\top \eta - a + b, \tag{6}$$

with primal feasibility, dual feasibility and complementary slackness. For crops i and j,

$$\mu_i - \mu_j = \lambda(d_i - d_j) + \beta(c_i - c_j) + (G^\top \eta)_i - (G^\top \eta)_j - (a_i - a_j) + (b_i - b_j). \tag{7}$$

The common land multiplier cancels, but budget, shared-constraint and bound terms remain. Equation 7 characterizes marginal optimality; it does not order acreage levels.

Dependence comparative statics require a distributional premise. If fixed marginals within a named family yield $L_{\theta_1}(x) \preceq_{\mathrm{cx}} L_{\theta_2}(x)$ for every x in a declared domain, then loss-CVaR is weakly ordered there. If the domain is all of X, risk-feasible sets are nested and optimal expected profit is weakly non-increasing. No crop-specific acreage conclusion follows without additional uniqueness and selection assumptions.

Accordingly, reversal is represented by sets. Let $g_{ij}^+(\theta)$ and $g_{ij}^-(\theta)$ be the maximum and minimum of $x_j - x_i$ over $S(\theta)$. Possible and universal reversal regions are the sets where g_{ij}^+ and g_{ij}^- exceed τ_x. Their boundaries may be empty, disconnected, multiple or interval-valued. A unique threshold requires conditions—including continuity, strict monotonicity and an endpoint crossing—that are not general.

Two teacher-Draft themes survive only with restricted meanings. Pseudo-diversification denotes preregistered discordance between ordinary correlation and lower-tail dependence; it implies neither inclusion nor exclusion and has no welfare content. For information, posterior-contingent value is non-negative when ignoring the signal remains feasible, and optimized value is weakly non-decreasing as action sets expand. The difference between informed and uninformed values need not itself be monotone, strictly positive or supermodular. Complete proofs and counterexample boundaries are provided in the Supplementary Information and the canonical theory package.

5 Numerical experiments

The numerical experiment probes the repaired propositions rather than calibrating a representative farmer. A preregistered mixed-factor design combines named copula and marginal families with Kendall dependence, a CVaR frontier quantile, budget tightness, a dominant-crop cap and a contract minimum. It contains 90 cells and five fixed replications per cell, yielding 450 primary optimizations. Parameters not identified by public aggregate data are labelled illustrative and are varied over frozen ranges.

Every primary solve reached an optimal status. Reverse-order independent replay passed for 450 replications, and all 9 solver-method comparisons passed. Direct CVaR evaluation, tail-weight identities

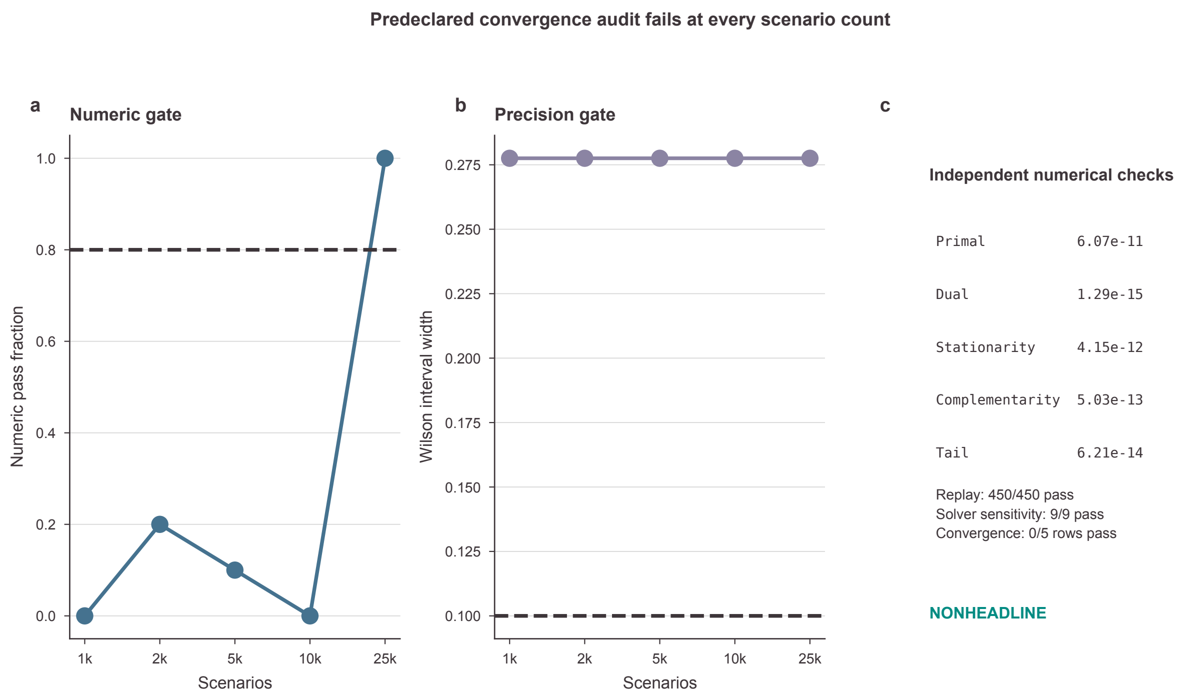

Figure 2: Predeclared convergence and numerical diagnostics. None of the five scenario-count rows passes both gates; the panel is supplementary and nonheadline.

and KKT residuals were checked against frozen tolerances. These results support implementation integrity, not the substantive prevalence of reversal.

The substantive landscape is deliberately supplementary. Universal reversal occurs in 150 replications; the CVaR constraint binds in 245 and is slack in 205; and the preregistered pseudo-diversification diagnostic is flagged in 84. These variations refute any blanket assertion that reversal requires a binding CVaR constraint or that dependence produces one monotone threshold.

Most importantly, the convergence audit fails. Across 5 scenario-count rows, only 0 satisfy the joint numeric-pass and interval-width rule; the largest Wilson interval width remains 0.278. The simulation figures and substantive counts are therefore marked NONHEADLINE. We report this adverse result because passing solver diagnostics and failing inferential precision are different facts.

6 Data and empirical design

The empirical analysis uses an official January 2025 US crop-production summary containing state planted acreage and yield for crop years 2022–2024. We retain state-years with complete acreage and yield for corn, soybeans and winter wheat, without imputing published missing codes. This produces 231 crop rows, 77 three-crop state-years and 26 states. The selected complete-case sample is not a nationally representative farm panel.

Four ranking definitions are frozen before comparison with acreage. Relative yield divides state yield by the same-year national yield. Standardized revenue multiplies state yield by a national same-year price. Operating margin subtracts national operating cost per planted acre, and total-cost margin subtracts national total cost. Monetary variables are expressed in real 2024 US dollars using CPI-U. The price and cost standardization is transparent accounting: it is neither a state realized margin nor a welfare measure.

Within each state-year and definition, crops receive descending score ranks and descending planted-acreage ranks. Pairwise inversions count discordant crop pairs. Top-rank reversal records whether the highest-scoring crop is not the acreage leader; strong reversal additionally requires the score leader to have less acreage than every lower-scoring crop under the frozen rule. Ties and sample sizes are retained rather than silently removed.

The primary estimand is descriptive prevalence, not a structural parameter. Exact enumeration of the six orderings of three crops provides a *combinatorial reference*: four of six permutations move the top-ranked crop from first place, yielding two-thirds. It is not a sampling null and no p-value is constructed from it. Year and state summaries describe heterogeneity but do not support causal geographic comparisons.

The empirical pipeline is isolated and reproducible. Two raw-to-analysis executions reproduce all processed and output artifacts byte for byte. Data-source identifiers, timestamps, raw checksums, processing contracts and parameter scope are recorded in the evidence registry. The admitted sources do not reveal private budgets, rotations, contracts, farmer risk limits or objectives; consequently, observed acreage cannot be classified as an optimum of the model.

7 Empirical results

Ranking and acreage orders often differ in the selected state sample, but the magnitude depends on how the ranking is constructed (Fig. 3). For operating margin, the top-ranked crop differs from the acreage leader in 63 of 77 state-years (81.8%). Strong reversal occurs in 41 state-years (53.2%). The result shows descriptive discordance between two orderings; it does not show that farmers ignored a feasible recommendation.

Definition sensitivity is substantial. Across relative yield, standardized revenue, operating margin and total-cost margin, top-rank reversal rates span 62.3% to 87.0%. Strong-reversal rates vary even more. Thus, there is no single empirical "ranking-reversal rate" independent of score construction. This sensitivity is substantively useful: the ordinal-to-cardinal gap is visible, but its measured prevalence inherits the definition of the ordinal object.

The operating-margin pattern is present in each sample year, with different rates and state counts. State cells also vary, and one state contributes only two complete years. These comparisons are descriptive because the panel is short and state composition is selected by three-crop completeness. The figure displays every cell's denominator so that a dark color cannot be mistaken for high precision.

The national aggregate provides an important null comparison. Real operating-margin and planted-acreage ranks agree in all 9 crop-year comparisons, so the number of national reversals is 0. This null does not invalidate state heterogeneity; it shows that aggregation level changes the empirical pattern and must be declared. It also cautions against extrapolating the state-sample rate to the country as a whole.

8 Robustness and extensions

Leave-one-year-out summaries retain the main qualitative ordering across all four definitions, but their movement is not interpreted as sampling uncertainty from a representative population. The exercise instead shows whether one year alone mechanically generates the full-sample rate. With only three crop years, it cannot establish temporal stability.

A leakage-free validation ranks 2024 relative yield using only the two earlier years. Among 25 complete states, 22 (88.0%) show top-rank disagreement with 2024 acreage. This is a low-powered descriptive validation, not a forecast-performance result: two training years cannot identify a production function or a stable predictive model.

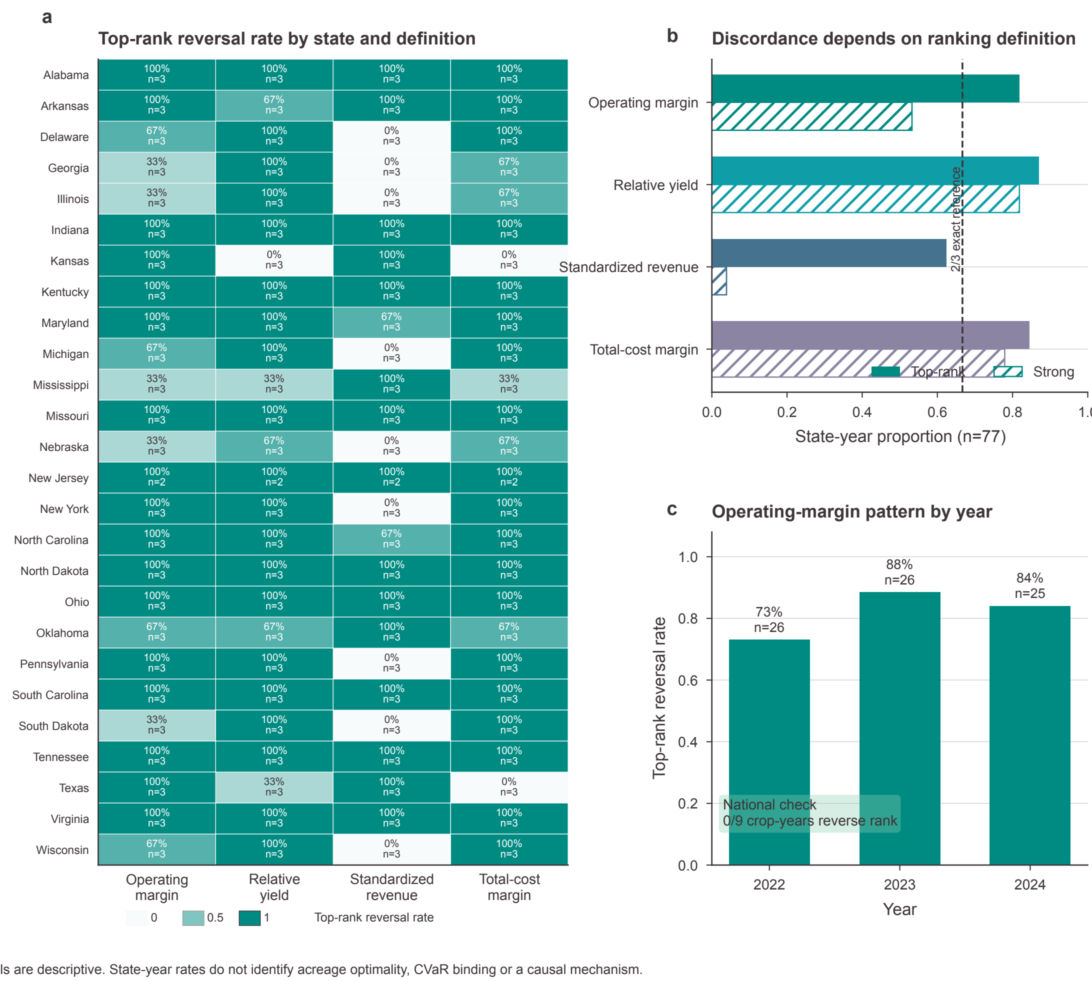

Figure 3: Descriptive ranking discordance across states and definitions. The heatmap reports rates and denominators; bars compare top-rank and strong reversal with the exact two-thirds combinatorial reference; yearly operating-margin rates are shown with the retained national null.

Robustness across ranking definitions is itself a boundary result. The operating-margin, relative-yield, standardized-revenue and total-cost-margin constructions answer different accounting questions, and none incorporates private feasibility. Agreement across definitions can strengthen confidence that a rank mismatch is not unique to one score, but disagreement cannot select the "correct" latent objective.

The framework admits several extensions that are not estimated here. Stochastic costs can enter per-acre profit; integer or fixed-charge decisions can replace the polytope; recourse can be introduced after a signal; and named dependence families can be compared under verified distributional orders. Such extensions require new assumptions and data. They cannot be inferred from the current acreage table.

Supplementary Fig. S3 combines the leave-one-year-out, lagged and national checks. Together they preserve three distinct outcomes: state-level descriptive discordance, a low-powered temporally ordered replication, and a national null. None identifies acreage optimality, a binding CVaR limit, copula causality, state downside risk or a welfare loss.

9 Discussion

The central contribution is a boundary on interpretation. A crop ranking can prioritize alternatives without determining the scale at which they should be used. Moving from rank to acreage requires cardinal payoffs, a feasible set, a risk convention and a rule for selecting among multiple optima. This distinction is easy to obscure because both objects can be displayed as ordered crop lists, yet they answer different questions.

The theory sharpens that distinction in three ways. It treats reversal as a property that may concern some, all or one selected optimizer; retains every operational dual term in the marginal condition; and replaces scalar dependence intuition with a restricted distributional order. Feasibility alone can force reversal, while a slack risk constraint can leave an unconstrained optimum unchanged. Consequently, observing discordance is insufficient to attribute it to downside risk or dependence.

The empirical evidence is consistent with the practical relevance of the distinction. State-level ranking and acreage orders often disagree in the admitted complete-case sample, and the pattern survives several definitions and temporal checks. At the same time, the wide spread across definitions and the national null show why one headline prevalence number would be misleading. The analysis documents a phenomenon to explain; it does not recover the explanation.

The failed convergence audit further disciplines interpretation. Numerical reproducibility establishes that code reruns and solvers agree, but it does not guarantee that finite-replication probabilities are precise. Keeping the simulation landscape nonheadline protects the main claim from an attractive but underpowered mechanism story. Future simulation work should increase independent replications under the frozen design or preregister a revised design before promoting prevalence or threshold claims.

Several limitations define the present scope. The state panel covers only 2022–2024 and includes states with complete observations for all three crops. National prices and costs standardize accounting but do not represent local realized margins. Observed acreage embeds unobserved rotations, contracts, expectations, policies and adjustment costs. Farmer-specific risk tolerance and feasible sets are unavailable. The optimizing mechanism is therefore not identified. The linear model omits fixed costs, endogenous prices, spatial interaction and multi-period rotation dynamics. These limits preclude causal, welfare and revealed-optimality interpretations.

The practical implication is therefore procedural. Ranking systems should be accompanied by the decision environment in which their recommendations will be used. At minimum, users need the cardinal score mapping, constraint provenance, uncertainty representation, selection rule and sensitivity to alternative definitions. Where these inputs are unavailable, a ranking-versus-acreage comparison should be labelled descriptive rather than converted into an efficiency or compliance judgment.

10 Conclusion

This study establishes that ordinal crop rankings do not, by themselves, determine constrained land allocations. A set-valued stochastic programme clarifies possible, universal and selected reversal, while repaired structural results show how feasibility, downside risk, dependence and optimizer multiplicity enter without implying a universal threshold. Official state data document substantial but definition-sensitive ranking–acreage discordance, alongside a national null. The simulation provides transparent diagnostics but remains nonheadline because its convergence rule fails.

The data identify descriptive ranking–acreage discordance, not its optimizing mechanism. Distinguishing operational, risk and dependence mechanisms requires longer panels, farm-level constraints and adequately replicated preregistered simulations. Rankings should therefore remain inputs to explicit decisions, not allocations in disguise.

11 Methods

11.1 Theory and optimization

We audited every definition and theorem in the preserved teacher Draft before implementation. The canonical model uses loss rather than the favourable profit tail, permits general finite-scenario weights, leaves the VaR auxiliary variable free, and retains land, budget, shared-constraint and bound multipliers. Possible and universal reversal are computed by minimizing and maximizing the relevant acreage gap over the objective-equivalent optimal face. All equivalence and feasibility comparisons use frozen numerical tolerances.

11.2 Scenario design and computation

The formal design was frozen before execution. Latin-hypercube cells and anchors vary only documented factors; each cell uses fixed seeds and a common scenario count, with a separate five-point convergence grid. Gaussian, Student-t and empirical-resampling marginals are paired with named Gaussian, Student-t and Clayton copulas. Calibration uses the admitted national crop-year panel, while budget, rotation, contract and risk-frontier stresses remain illustrative. Each solve records primal values, named slacks, binding flags, duals, tail weights, KKT residuals and an optimal-face audit. Independent replay reverses task order, and selected anchors are rerun with multiple HiGHS methods.

11.3 Official data and ranking construction

Raw sources are admitted only through a registry containing authority, stable identifier, retrieval date, geography, time support, units and checksum. The empirical parser reads the frozen official crop-summary snapshot and applies a complete three-crop rule. Same-year national prices and costs are joined only after crop and year reconciliation and are converted to real 2024 dollars. Four ranking definitions are computed without post-result selection. State-year outcomes, sample flow, heterogeneity, definition sensitivity, lagged validation and national checks are written to tidy CSV files.

11.4 Reproducibility and claim governance

Theory, data, simulation, empirical and visualization validators run before the repository manifest is rebuilt. Figure source data are deterministic extracts of verified outputs, and vector/raster exports have independent checksum ledgers. Manuscript numbers are generated by `scripts/generate_manuscript_inputs.py`; the number registry records each macro's output file and field. Claim–citation and figure/table registries restrict prose to verified sources and evidence levels. Full protocols, proofs, additional tables and robustness figures are in the Supplementary Information.

Declarations

Acknowledgements. To be completed by the authors before submission.

Author contributions. To be completed using the journal's CRediT format.

Competing interests. The authors must confirm the final statement before submission.

Ethics. The study uses public aggregate data and no human-participant or animal data.

Data availability. All admitted source identifiers, immutable raw-file checksums, processing contracts and derived analysis files are recorded in the repository's data and evidence registries. Public-source redistribution remains subject to the originating agencies' terms.

Code availability. The complete theory, optimization, simulation, empirical, visualization and manuscript pipelines are versioned in the companion repository. Reproduction commands and checksums are provided with each artifact class.

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